



Science — Cbse Half-Yearly Exam — 80 Marks

Subject: Science | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	Which of the following correctly pairs a connective tissue with its connection? (A) Tendon - connects bone to bone (B) Ligament - connects muscle to bone (C) Cartilage - connects bone to muscle (D) Tendon - connects muscle to bone <i>[Ch 3: Tissues in Action]</i>	1
2	If a plant is unable to transport food from its leaves to its roots, which specific tissue is most likely malfunctioning? (A) Xylem (B) Epidermis (C) Phloem (D) Sclerenchyma <i>[Ch 3: Tissues in Action]</i>	1
3	In which year did Isaac Newton present his three laws of motion? (A) 1607 (B) 1687 (C) 1707 (D) 1757 <i>[Ch 6: How Forces Affect Motion]</i>	1



Q.NO.	QUESTIONS	MARKS
4	Which of the following is classified as a water-in-oil emulsion? (A) Milk (B) Butter (C) Vanishing cream (D) Sugar solution <i>[Ch 5: Exploring Mixtures and their Separation]</i>	1
5	A sound wave of frequency 20 kHz propagates in air with speed 344 m s^{-1}. What is its wavelength? (A) 17.2 m (B) 17.2 cm (C) 1.72 cm (D) 0.172 m <i>[Ch 10: Sound Waves: Characteristics and Applications]</i>	1
6	Assertion (A): In the vacuum bell jar experiment, the sound of an electric bell becomes fainter as air is sucked out from the bell jar. Reason (R): Sound needs a medium to propagate and cannot travel through vacuum. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 10: Sound Waves: Characteristics and Applications]</i>	1
7	Imagine an object moving on a perfectly smooth horizontal floor where the force of friction between them is zero. What will happen to the object's velocity? (A) It will not decrease and the object will continue moving forever. (B) It will gradually decrease, but the object will never stop. (C) It will decrease at the same rate as on a rough surface. (D) It will instantly drop to zero. <i>[Ch 6: How Forces Affect Motion]</i>	1
8	Assertion (A): Cultured phloem cells of carrot in a liquid medium with nutrients show a 20% increase in fresh weight when supplied with air. Reason (R): The presence of light along with a solid medium and nutrients increases the biomass of the cultured plant cells. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 3: Tissues in Action]</i>	1



Q.NO.	QUESTIONS	MARKS
9	What is the SI unit of force and how is its symbol correctly written? (A) newton, written as N (B) Newton, written as n (C) newton, written as n (D) Newton, written as N <i>[Ch 6: How Forces Affect Motion]</i>	1
10	Sound is a longitudinal mechanical wave that: (A) Can travel through vacuum (B) Does not require a medium to propagate (C) Only travels through gases (D) Needs a medium to travel <i>[Ch 10: Sound Waves: Characteristics and Applications]</i>	1
11	What is a likely consequence of errors in meiosis? (A) Genetic disorders and reduced fertility (B) Formation of tumours in body cells (C) Uncontrolled cell divisions (D) Abnormal number of chromosomes only in skin cells <i>[Ch 2: Cell: The Building Block of Life]</i>	1
12	Assertion (A): The path of a laser beam is not visible when passed through a true solution. Reason (R): The particles in a suspension scatter the beam of light, making its path visible. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 5: Exploring Mixtures and their Separation]</i>	1
13	Assertion (A): The experiment conducted by J. Craig Venter and his team created a completely new cell from scratch. Reason (R): The synthetic DNA inserted into the cell controlled the structure and activities of the cell. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 2: Cell: The Building Block of Life]</i>	1



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14	What is the limit of resolution of the human eye when viewing from a distance of about 25 cm? (A) 0.01 mm (B) 0.1 mm (C) 1.0 mm (D) 10 mm <i>[Ch 2: Cell: The Building Block of Life]</i>	1
15	A solid that is made up of particles arranged in a regular geometric pattern is called: (A) An amorphous solid (B) A precipitate (C) A crystal (D) An alloy <i>[Ch 5: Exploring Mixtures and their Separation]</i>	1
16	For an object moving in a straight line in the same direction, when is its acceleration considered constant? (A) When its velocity is zero (B) When it travels equal distances in equal intervals of time (C) When its displacement is zero (D) When the magnitude of its velocity increases or decreases by equal amounts in equal intervals of time <i>[Ch 4: Describing Motion Around Us]</i>	1
17	If an object moving in a straight line travels equal distances in equal intervals of time (for all possible choices of time intervals), its motion is described as: (A) Uniform motion (B) Non-uniform motion (C) Accelerated motion (D) Oscillatory motion <i>[Ch 4: Describing Motion Around Us]</i>	1
18	Why is blood fluid while bone is hard? (A) The matrix in blood is watery and soft, while in bones it is hard and rigid. (B) Blood lacks a matrix, while bone has a rigid matrix. (C) Blood contains only cells, while bones contain only fibres. (D) The matrix in blood is hard, while in bones it is jelly-like. <i>[Ch 3: Tissues in Action]</i>	1



Q.NO.	QUESTIONS	MARKS
19	Stainless steel is an alloy of iron that contains which of the following elements? (A) Only carbon and chromium (B) Copper (80%) and zinc (20%) (C) Copper (80%) and tin (20%) (D) Carbon (about 0.03–0.8%), chromium (16–18%), nickel (10.0–14.0%), and molybdenum (2.0–3.0%) <i>[Ch 5: Exploring Mixtures and their Separation]</i>	1
20	Assertion (A): When a marble moving along the inner boundary of a ring is released by lifting the ring, it continues to move in a straight line. Reason (R): The velocity at a point on the circular path is along the tangent to the circle at that point, and the marble continues to move in the direction it has been moving at the instant of release. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 4: Describing Motion Around Us]</i>	1
Section B Attempt 5 out of 7.		
21	State the significance of mitosis in multicellular organisms. <i>[Ch 2: Cell: The Building Block of Life]</i>	2
	OR Differentiate between sublimation and deposition based on the state changes involved. <i>[Ch 5: Exploring Mixtures and their Separation]</i>	
22	A naval sonar signal sent into seawater returns after 0.90 s. If the speed of sound in seawater is 1530 m s^{-1}, how far is the object? <i>[Ch 10: Sound Waves: Characteristics and Applications]</i>	2
	OR A fielder catching a fast-moving cricket ball gradually pulls their hands backwards. Explain how this reduces the chance of injury. <i>[Ch 6: How Forces Affect Motion]</i>	
23	What is the thickness of the cell membrane and what is it made of? <i>[Ch 2: Cell: The Building Block of Life]</i>	2
24	What does a straight line parallel to the time axis on a position-time graph indicate about the nature of motion of the object? <i>[Ch 4: Describing Motion Around Us]</i>	2



Q.NO.	QUESTIONS	MARKS
25	What is the primary function of tendons in the body, and how do they interact with the muscular and skeletal systems? <i>[Ch 3: Tissues in Action]</i>	2
Section C Attempt 5 out of 7.		
26	How can a spring balance be used to compare the force of friction on a wooden block across different surfaces? What does a smaller reading indicate? <i>[Ch 6: How Forces Affect Motion]</i>	3
	OR How does the epidermis protect plants from various environmental factors, and what role does the cuticle play in plants living in dry habitats? <i>[Ch 3: Tissues in Action]</i>	
27	What is pitch? How does the pitch of a sound relate to its frequency, and what changes occur in the vocal cords of boys during adolescence that affect their voice pitch? <i>[Ch 10: Sound Waves: Characteristics and Applications]</i>	3
	OR Sewage water treatment involves several steps to clean the water before it is released into the environment. Explain the role of coagulation in this process and state one way the clean water obtained can be reused. <i>[Ch 5: Exploring Mixtures and their Separation]</i>	
28	A car starts from rest and its velocity reaches 24 m s^{-1} in 6 s. Find the average acceleration and the distance travelled in these 6 s. <i>[Ch 4: Describing Motion Around Us]</i>	3
29	Differentiate between the following pairs of terms based on the clues given in parentheses: (i) Cell membrane and cell wall (permeability) (ii) RER and SER (structure) (iii) Chloroplasts and chromoplasts (pigments) <i>[Ch 2: Cell: The Building Block of Life]</i>	3
30	What is Programmed Cell Death (PCD)? Explain its role in human embryonic development. <i>[Ch 2: Cell: The Building Block of Life]</i>	3
Section D Attempt 4 out of 6.		



Q.NO.	QUESTIONS	MARKS
31	Describe the step-by-step procedure for plotting a position-time graph from given data. In your explanation, include how to draw the axes, choose suitable scales, plot the data points, and connect them to form the graph. <i>[Ch 4: Describing Motion Around Us]</i>	5
	OR What are emulsions? Differentiate between oil-in-water and water-in-oil emulsions with examples. Explain the role of emulsifying agents and describe how you can prepare an emulsion using cooking oil and water. <i>[Ch 5: Exploring Mixtures and their Separation]</i>	
32	Describe the vacuum bell jar experiment to demonstrate that sound needs a medium to propagate. What observations are made when air is removed and then let back in? Also, explain how this principle applies to astronauts on spacewalks. <i>[Ch 10: Sound Waves: Characteristics and Applications]</i>	5
	OR Explain the process of osmosis and how it differs from diffusion. Describe the behaviour of cells when placed in isotonic, hypotonic, and hypertonic solutions, using the example of potato pieces in plain water and salt solution. <i>[Ch 2: Cell: The Building Block of Life]</i>	
33	Describe how the force of friction affects the motion of an object on different surfaces. Explain the investigation using a stack of coins and a rubber band on surfaces such as a wooden table top, a laminated table top, and a polished marble floor, and state the conclusions drawn about the relationship between friction and the distance travelled. <i>[Ch 6: How Forces Affect Motion]</i>	5
34	Define totipotency. Describe how F.C. Steward demonstrated this property using carrot phloem cells, and explain the processes of dedifferentiation and redifferentiation that occur during the regeneration of a complete plant. <i>[Ch 3: Tissues in Action]</i>	5
Section E Attempt 3 out of 4.		



Q.NO.	QUESTIONS	MARKS
35	Case Study: Plant Protective and Supporting Tissues A botanist comparing plants from different habitats observes that desert plants have a thick waxy coating on their surface, while aquatic plants float easily on water. She notices that young plant stems bend without breaking, but the hard shells of walnuts are extremely tough and cannot bend. She examines the transverse section of a sunflower stem under a microscope and identifies several types of permanent tissues. She finds that permanent tissues can be simple (composed of only one type of cell) or complex (composed of more than one type of cell), and each tissue is specialised to perform a specific function such as support, transport, or storage. (i) What is the waxy layer of cutin on the epidermis called? (ii) Which type of simple permanent tissue allows plant stems and tendrils to bend without breaking? Name the chemical responsible for this flexibility. (iii) A walnut shell is hard and strong. Identify the tissue, the chemical deposited in its cell walls, and state whether its cells are living or dead. (iv) How does the specialised parenchyma in aquatic plants help them survive in their habitat? <i>[Ch 3: Tissues in Action]</i>	5
	OR Case Study: Concentration of Solutions in Everyday Products A student examines three products. A packaged food item contains 8 g of salt in 200 g of the product. A glucose solution bottle from a pharmacy is labelled 5% m/v, meaning 5 g of glucose is present in 100 mL of the solution. A milk powder packet lists its protein content as 15% m/m. The student notes that mass by mass percentage is commonly used for heterogeneous mixtures like milk powder, while mass by volume percentage is used in medicines and laboratories where measuring the volume of a liquid is easier than weighing it. (i) What does 5% m/v on the glucose solution bottle indicate? (ii) Why is mass by volume percentage used for medicines like glucose solution instead of mass by mass percentage? (iii) Calculate the mass by mass percentage of salt in the packaged food item if it contains 8 g of salt in 200 g of the product. (iv) Milk powder is labelled using mass by mass percentage for its composition. What type of mixture is milk powder, and why is this method of expressing concentration appropriate for it? <i>[Ch 5: Exploring Mixtures and their Separation]</i>	



Q.NO.	QUESTIONS	MARKS
36	Case Study: Average Acceleration When a bus moves on a long straight highway, its velocity can change. The average acceleration of the bus over a time interval is the change in its velocity divided by the time interval. For instance, if a driver presses the accelerator, the velocity increases, and the average acceleration is in the direction of velocity. Conversely, if the driver presses the brake, the velocity decreases, and the average acceleration is opposite to the direction of velocity. Interestingly, a bus moving at a constant velocity, even if it is very fast, has zero acceleration because acceleration depends on how quickly velocity changes. (i) What is the SI unit of average acceleration? (ii) A bus moves on a straight highway at a constant velocity. What is its average acceleration? (iii) A bus increases its velocity from 10 m s^{-1} to 15 m s^{-1} in a time interval of 10 s. Calculate the average acceleration. (iv) When the magnitude of velocity of a bus is decreasing, in which direction does the average acceleration act relative to the velocity? <i>[Ch 4: Describing Motion Around Us]</i>	5
37	Case Study: Energy and Food Synthesis in Cells Plants and animals have specialised organelles to meet their energy and nutritional needs. In animal cells, the 'powerhouses' generate energy through cellular respiration, breaking down glucose to produce ATP. These organelles have an outer smooth, porous membrane and an inner membrane folded into cristae. Meanwhile, plant cells contain double-membrane-bound organelles that synthesise food using sunlight. Inside these organelles, a semi-fluid substance called the stroma contains disc-shaped structures that hold the green pigment chlorophyll. Interestingly, both types of organelles possess their own DNA and ribosomes, allowing them to synthesise some of their own proteins. These features suggest they share an evolutionary history with single-celled organisms. (i) What is the function of the cristae present in the inner membrane of mitochondria? (ii) Why is ATP referred to as the 'energy currency' of the cell? (iii) Describe the role of the stroma and the disc-shaped membrane structures inside the chloroplast during photosynthesis. (iv) Based on their characteristics, what can be inferred about the evolutionary history of mitochondria and plastids? <i>[Ch 2: Cell: The Building Block of Life]</i>	5

